# MATH 341 — Differential Equations

**Methods & Techniques Quick Reference**

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> **Note**
>
> **Textbook:** Logan, J.D. *A First Course in Differential Equations*, 3rd ed. (Springer, 2015).
> **Sections covered:** All sections except §2.6.
> **Course website:** https://math341diffeq.netlify.app/

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# 1 First-Order Equations (Chapter 1)

## 1.1 Quick Decision Guide

| Form of Equation | Technique | See Section |
|---|---|---|
| $y' = f(x)\,g(y)$ | Separation of Variables | §1.2 |
| $y' + p(x)\,y = q(x)$ | Integrating Factor | §1.3 |
| $M\,dx + N\,dy = 0$, $M_y = N_x$ | Exact Equation | §1.4 |
| $y' = f(y/x)$ | Substitution $v = y/x$ | §1.5 |
| $y' = f(ax + by + c)$ | Substitution $v = ax + by + c$ | §1.5 |
| $y' + p(x)\,y = q(x)\,y^n$ | Bernoulli Substitution | §1.5 |
| None of the above | Numerical Methods (§1.6) or Qualitative Analysis | §1.1 |

---

## 1.2 Separation of Variables (§1.2)

**When to use:** The right-hand side factors as $f(x) \cdot g(y)$.

**Form:** $\dfrac{dy}{dx} = f(x)\, g(y)$

**Steps:**

1. Rewrite as $\dfrac{dy}{g(y)} = f(x)\, dx$ (check $g(y) \neq 0$).
2. Integrate both sides: $\displaystyle\int \frac{dy}{g(y)} = \int f(x)\, dx + C$.
3. Solve for $y$ (if possible); apply initial condition to find $C$.

> **Warning**
>
> **Watch out:** Always check for **equilibrium solutions** where $g(y) = 0$ — these may be lost when dividing.

**Example:** $y' = xy$
$\Rightarrow \dfrac{dy}{y} = x\, dx \Rightarrow \ln|y| = \dfrac{x^2}{2} + C \Rightarrow y = Ae^{x^2/2}$

---

## 1.3 Integrating Factor / Linear First-Order (§1.3)

**When to use:** Equation is linear in $y$ (or can be put in standard linear form).

**Standard form:** $y' + p(x)\, y = q(x)$

**Steps:**

1. Identify $p(x)$ and $q(x)$.
2. Compute integrating factor: $\mu(x) = e^{\int p(x)\, dx}$.
3. Multiply both sides by $\mu$: $(\mu\, y)' = \mu\, q(x)$.
4. Integrate: $\mu\, y = \displaystyle\int \mu\, q(x)\, dx + C$.
5. Solve: $y = \dfrac{1}{\mu}\left(\displaystyle\int \mu\, q\, dx + C\right)$.

> **Tip**
>
> **Tip:** The general solution is $y = y_h + y_p$ where $y_h = C/\mu$ is the homogeneous solution and $y_p = \frac{1}{\mu}\int \mu q\, dx$ is a particular solution.

**Example:** $y' - 2xy = x$
$p(x) = -2x$, $\mu = e^{-x^2}$, then $(e^{-x^2} y)' = xe^{-x^2}$, giving $y = -\frac{1}{2} + Ce^{x^2}$.

---

## 1.4 Exact Equations (§1.4)

**When to use:** Equation written as $M(x,y)\,dx + N(x,y)\,dy = 0$ with $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$.

**Steps:**

1. Check exactness: $M_y = N_x$?
2. Find potential function $F(x,y)$ such that $F_x = M$, $F_y = N$.
3. Integrate $M$ w.r.t. $x$: $F = \displaystyle\int M\,dx + g(y)$.
4. Differentiate $F$ w.r.t. $y$ and set equal to $N$ to find $g'(y)$; integrate.
5. Solution: $F(x,y) = C$.

> **Warning**
>
> **Not exact?** Try integrating factors $\mu(x)$ or $\mu(y)$ to make it exact.
> If $\dfrac{M_y - N_x}{N}$ depends only on $x$, then $\mu = e^{\int \frac{M_y - N_x}{N}\,dx}$.

---

## 1.5 Substitution Methods (§1.5)

### 1.5.1 Homogeneous Equations

**Form:** $y' = f\left(\dfrac{y}{x}\right)$
**Substitution:** $v = y/x$, so $y = vx$ and $y' = v + xv'$.
Reduces to a **separable** equation in $v$ and $x$.

### 1.5.2 Bernoulli Equations

**Form:** $y' + p(x)\,y = q(x)\,y^n$, $n \neq 0, 1$
**Substitution:** $w = y^{1-n}$, so $w' = (1-n)y^{-n}y'$.
Reduces to a **linear** equation in $w$.

### 1.5.3 Linear Substitution

**Form:** $y' = f(ax + by + c)$
**Substitution:** $v = ax + by + c$, so $v' = a + by'$.
Reduces to a **separable** equation.

---

## 1.6 Autonomous Equations & Qualitative Analysis (§1.1, §1.6)

**Form:** $y' = f(y)$ (right side depends only on $y$)

**Key concepts:**

- **Equilibria (critical points):** Solve $f(y^*) = 0$.
- **Stability:**
  - $f'(y^*) < 0$: **stable** (attractor)
  - $f'(y^*) > 0$: **unstable** (repeller)
- **Phase line:** Plot $f(y)$ vs. $y$; arrows point right where $f > 0$, left where $f < 0$.

> **Tip**
>
> **Tip:** The phase line immediately reveals long-term behavior without solving the ODE explicitly.

---

## 1.7 Numerical Methods (§1.7)

### 1.7.1 Euler's Method

**Formula:** $y_{n+1} = y_n + h\, f(x_n, y_n)$, where $h$ is the step size.

**Error:** $O(h)$ per step — first-order method. Use small $h$ for accuracy.

### 1.7.2 Improved Euler (Heun's Method)

**Predictor:** $\tilde{y}_{n+1} = y_n + h\, f(x_n, y_n)$

**Corrector:** $y_{n+1} = y_n + \dfrac{h}{2}[f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1})]$

**Error:** $O(h^2)$ — second-order method.

### 1.7.3 Runge-Kutta (RK4)

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(x_n, y_n), \quad k_2 = f\left(x_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_1\right)$$

$$k_3 = f\left(x_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_2\right), \quad k_4 = f(x_n + h, y_n + hk_3)$$

**Error:** $O(h^4)$ — standard workhorse for numerical ODEs.

---

# 2 Second-Order Linear Equations (Chapter 2)

## 2.1 Quick Decision Guide

| Equation Type | Technique | See Section |
|---|---|---|
| $ay'' + by' + cy = 0$ | Characteristic Equation | §2.1 |
| $ay'' + by' + cy = g(t)$, $g =$ poly/exp/sin/cos | Undetermined Coefficients | §2.3 |
| $ay'' + by' + cy = g(t)$, $g =$ other | Variation of Parameters | §2.4 |
| $x^2y'' + xy' + \alpha y = 0$ | Cauchy–Euler Equation | §2.5 |
| Two solutions $y_1, y_2$ needed | Reduction of Order | §2.2 |

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## 2.2 Homogeneous Linear Equations: Characteristic Equation (§2.1)

**Standard form:** $ay'' + by' + cy = 0$

**Characteristic equation:** $ar^2 + br + c = 0$, with roots $r = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

| Discriminant | Roots | General Solution |
|---|---|---|
| $b^2 - 4ac > 0$ | Two real: $r_1 \neq r_2$ | $y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ |
| $b^2 - 4ac = 0$ | Repeated: $r_1 = r_2 = r$ | $y = (C_1 + C_2 t)\, e^{rt}$ |
| $b^2 - 4ac < 0$ | Complex: $r = \alpha \pm \beta i$ | $y = e^{\alpha t}(C_1 \cos \beta t + C_2 \sin \beta t)$ |

> **Tip**
>
> For complex roots $r = \alpha \pm \beta i$: $\alpha = -b/(2a)$, $\beta = \sqrt{4ac - b^2}/(2a)$.

---

## 2.3 Reduction of Order (§2.2)

**When to use:** One solution $y_1$ is known; find a second linearly independent solution $y_2$.

**Substitution:** $y_2 = v(t)\, y_1(t)$

**Steps:**

1. Substitute $y = v\, y_1$ into the ODE.
2. Use $y_1'' + p(t)y_1' + q(t)y_1 = 0$ to eliminate terms.
3. Let $w = v'$ to get a first-order linear ODE in $w$.

4. Solve for $w$, then integrate to find $v$, then $y_2 = v\,y_1$.

**Wronskian:** $W(y_1, y_2) = y_1 y_2' - y_1' y_2 \neq 0$ confirms linear independence.

**Abel's Formula:** $W(t) = W(t_0)\, e^{-\int p(t)\,dt}$

---

## 2.4 Undetermined Coefficients (§2.3)

**When to use:** $ay'' + by' + cy = g(t)$ where $g(t)$ is a combination of polynomials, exponentials, sines, or cosines.

**General solution:** $y = y_h + y_p$

### 2.4.1 Particular Solution Guesses

| Forcing $g(t)$ | Guess for $y_p$ |
|---|---|
| $P_n(t)$ (degree-$n$ poly) | $A_n t^n + \cdots + A_1 t + A_0$ |
| $e^{\alpha t}$ | $Ae^{\alpha t}$ |
| $\sin(\beta t)$ or $\cos(\beta t)$ | $A\cos(\beta t) + B\sin(\beta t)$ |
| $e^{\alpha t}\sin(\beta t)$ or $e^{\alpha t}\cos(\beta t)$ | $e^{\alpha t}(A\cos\beta t + B\sin\beta t)$ |
| $P_n(t)\, e^{\alpha t}$ | $(A_n t^n + \cdots + A_0)\, e^{\alpha t}$ |

> **Warning**
>
> **Modification Rule:** If any term of the guess solves the *homogeneous* equation, multiply the entire guess by $t$ (or $t^2$ if needed for repeated roots).

---

## 2.5 Variation of Parameters (§2.4)

**When to use:** $y'' + p(t)y' + q(t)y = g(t)$ when $g(t)$ is not of the "undetermined coefficients" type, or the coefficients are non-constant.

**Requires:** Two known linearly independent solutions $y_1$, $y_2$ of the homogeneous equation.

**Formulas:**

$$y_p = -y_1 \int \frac{y_2\, g}{W}\, dt + y_2 \int \frac{y_1\, g}{W}\, dt$$

where $W = W(y_1, y_2) = y_1 y_2' - y_1' y_2$.

> **Note**
>
> Always write the ODE in **standard form** (leading coefficient 1) before applying this formula.

---

## 2.6 Cauchy–Euler Equations (§2.5)

**Form:** $x^2y'' + \alpha xy' + \beta y = 0$, $x > 0$

**Substitution:** $x = e^t$ (or try $y = x^r$)

**Characteristic equation:** $r(r-1) + \alpha r + \beta = 0 \Rightarrow r^2 + (\alpha - 1)r + \beta = 0$

| Roots | General Solution |
|---|---|
| Two real: $r_1 \neq r_2$ | $y = C_1 x^{r_1} + C_2 x^{r_2}$ |
| Repeated: $r_1 = r_2 = r$ | $y = x^r(C_1 + C_2 \ln x)$ |
| Complex: $r = \lambda \pm \mu i$ | $y = x^\lambda(C_1 \cos(\mu \ln x) + C_2 \sin(\mu \ln x))$ |

---

## 2.7 Applications: Spring-Mass System (§2.1–§2.3)

**Equation of motion:** $my'' + \gamma y' + ky = F(t)$

| Parameter | Meaning |
|---|---|
| $m > 0$ | Mass |
| $\gamma \geq 0$ | Damping coefficient |
| $k > 0$ | Spring constant |
| $F(t)$ | External forcing |

**Damping regimes** (unforced, $F = 0$):

| Condition | Regime | Behavior |
|---|---|---|
| $\gamma^2 - 4mk > 0$ | Overdamped | Exponential decay, no oscillation |
| $\gamma^2 - 4mk = 0$ | Critically damped | Fastest decay without oscillation |
| $\gamma^2 - 4mk < 0$ | Underdamped | Oscillates with decaying amplitude |
| $\gamma = 0$ | Undamped | Pure oscillation at $\omega_0 = \sqrt{k/m}$ |

> ⚠ Warning
>
> **Resonance:** Occurs when $F(t) = F_0 \cos(\omega_0 t)$ and $\gamma = 0$. The amplitude grows without bound — use the **Modification Rule** (multiply guess by $t$).

---

# 3 Laplace Transforms (Chapter 3)

## 3.1 Definition & Key Properties

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt, \quad s > 0$$

| Property | Formula |
|---|---|
| Linearity | $\mathcal{L}\{af + bg\} = aF(s) + bG(s)$ |
| Derivative | $\mathcal{L}\{f'\} = sF(s) - f(0)$ |
| Derivative | $\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$ |
| $n$-th Derivative | $\mathcal{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$ |
| $s$-shifting | $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ |
| $t$-shifting | $\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s)$ |
| Convolution | $\mathcal{L}\{f * g\} = F(s)\,G(s)$ |
| Multiplication by $t$ | $\mathcal{L}\{t\,f(t)\} = -F'(s)$ |

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## 3.2 Common Laplace Transform Table

| $f(t)$ | $F(s) = \mathcal{L}\{f\}$ | Condition |
|---|---|---|
| $1$ | $\dfrac{1}{s}$ | $s > 0$ |
| $t^n$ | $\dfrac{n!}{s^{n+1}}$ | $s > 0$ |
| $e^{at}$ | $\dfrac{1}{s-a}$ | $s > a$ |
| $\sin(bt)$ | $\dfrac{b}{s^2+b^2}$ | $s > 0$ |
| $\cos(bt)$ | $\dfrac{s}{s^2+b^2}$ | $s > 0$ |
| $e^{at}\sin(bt)$ | $\dfrac{b}{(s-a)^2+b^2}$ | $s > a$ |
| $e^{at}\cos(bt)$ | $\dfrac{s-a}{(s-a)^2+b^2}$ | $s > a$ |

| $f(t)$ | $F(s) = \mathcal{L}\{f\}$ | Condition |
|---|---|---|
| $t^n e^{at}$ | $\dfrac{n!}{(s-a)^{n+1}}$ | $s > a$ |
| $u_c(t)$ (unit step at $c$) | $\dfrac{e^{-cs}}{s}$ | $s > 0$ |
| $\delta(t-c)$ (Dirac delta) | $e^{-cs}$ | |

---

### 3.3 Solving IVPs with Laplace Transforms (§3.1–§3.3)

**When to use:** Constant-coefficient ODEs (any order), especially with discontinuous/impulsive forcing.

**Steps:**

1. Take $\mathcal{L}$ of both sides; use initial conditions automatically.
2. Solve algebraically for $Y(s) = \mathcal{L}\{y(t)\}$.
3. Use **partial fractions** to decompose $Y(s)$.
4. Apply $\mathcal{L}^{-1}$ (use table) to find $y(t)$.

> **Note**
>
> **Partial Fractions Reminder:**
>
> - Distinct linear factors: $\dfrac{A}{s-a} + \dfrac{B}{s-b}$
> - Repeated factor: $\dfrac{A}{s-a} + \dfrac{B}{(s-a)^2}$
> - Irreducible quadratic: $\dfrac{As+B}{s^2+b^2}$ — complete the square if needed

---

### 3.4 Step Functions & Piecewise Forcing (§3.2)

**Heaviside (unit step) function:**

$$u_c(t) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}$$

**Writing piecewise functions:** Any piecewise function can be written using $u_c$:

$$f(t) = \begin{cases} g(t), & t < c \\ h(t), & t \geq c \end{cases} = g(t) + [h(t) - g(t)]\, u_c(t)$$

**Key formula:** $\mathcal{L}\{u_c(t)\, f(t-c)\} = e^{-cs}F(s)$
(shift the function, then multiply by $e^{-cs}$)

---

### 3.5 Impulse Functions & Dirac Delta (§3.3)

**Dirac delta:** $\delta(t-c)$ models an instantaneous impulse at $t = c$.

$$\mathcal{L}\{\delta(t-c)\} = e^{-cs}, \quad c \geq 0$$

$$\int_{-\infty}^{\infty} f(t)\, \delta(t-c)\, dt = f(c)$$

**Physical meaning:** An impulsive force (hammer blow, lightning strike) applied at time $t = c$.

---

### 3.6 Convolution (§3.4)

**Definition:** $(f * g)(t) = \displaystyle\int_0^t f(t-\tau)\, g(\tau)\, d\tau$

**Convolution Theorem:** $\mathcal{L}\{f * g\} = F(s)\, G(s)$

**When to use:** Invert a product $F(s)G(s)$ when partial fractions are unwieldy, or express solutions to IVPs with general forcing as a convolution integral.

---

## 4 Linear Systems (Chapter 4)

### 4.1 System Setup

A **first-order linear system:**

$$\mathbf{x}' = A\mathbf{x} + \mathbf{b}(t), \quad \mathbf{x} \in \mathbb{R}^n$$

For a homogeneous system ($\mathbf{b} = \mathbf{0}$), seek solutions $\mathbf{x} = \mathbf{v}\, e^{\lambda t}$, giving the **eigenvalue problem** $A\mathbf{v} = \lambda\mathbf{v}$.

### 4.2 Quick Decision Guide: $2 \times 2$ Systems

| Eigenvalues of $A$ | Solution Form | Phase Portrait |
|---|---|---|
| Real, distinct, same sign | $C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$ | Node (stable if $\lambda < 0$, unstable if $\lambda > 0$) |
| Real, distinct, opposite sign | $C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$ | Saddle (always unstable) |
| Repeated real $\lambda$ | See below | Improper node |
| Complex $\lambda = \alpha \pm \beta i$ | $e^{\alpha t}(\dots \cos\beta t + \dots \sin\beta t)$ | Spiral or center |

---

## 4.3 Solving $2 \times 2$ Homogeneous Systems (§4.1–§4.2)

**Steps:**

1. Find eigenvalues: $\det(A - \lambda I) = 0 \Rightarrow \lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$.
2. For each $\lambda$, find eigenvector $\mathbf{v}$: $(A - \lambda I)\mathbf{v} = \mathbf{0}$.
3. Write general solution.

### 4.3.1 Case 1: Distinct Real Eigenvalues

$$\mathbf{x}(t) = C_1\mathbf{v}_1e^{\lambda_1 t} + C_2\mathbf{v}_2e^{\lambda_2 t}$$

### 4.3.2 Case 2: Complex Eigenvalues $\lambda = \alpha \pm \beta i$

Compute one complex solution $\mathbf{x}_1 = e^{\alpha t}(\mathbf{p}\cos\beta t - \mathbf{q}\sin\beta t)$ using the real and imaginary parts of the complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$:

$$\mathbf{x}(t) = C_1e^{\alpha t}(\mathbf{p}\cos\beta t - \mathbf{q}\sin\beta t) + C_2e^{\alpha t}(\mathbf{p}\sin\beta t + \mathbf{q}\cos\beta t)$$

### 4.3.3 Case 3: Repeated Eigenvalue $\lambda$ (defective)

One eigenvector $\mathbf{v}$; find **generalized eigenvector** $\mathbf{w}$: $(A - \lambda I)\mathbf{w} = \mathbf{v}$.

$$\mathbf{x}(t) = C_1\mathbf{v}\,e^{\lambda t} + C_2(\mathbf{v}\,t + \mathbf{w})\,e^{\lambda t}$$

---

## 4.4 Phase Plane Analysis (§4.3)

**Trace-Determinant Plane:** For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, let $\tau = \mathrm{tr}(A) = a + d$ and $\Delta = \det(A) = ad - bc$.

| Region | Eigenvalue Type | Equilibrium |
|---|---|---|
| $\Delta < 0$ | Real, opposite signs | Saddle (unstable) |
| $\Delta > 0$, $\tau^2 - 4\Delta > 0$, $\tau < 0$ | Real, both negative | Stable node |
| $\Delta > 0$, $\tau^2 - 4\Delta > 0$, $\tau > 0$ | Real, both positive | Unstable node |
| $\Delta > 0$, $\tau^2 - 4\Delta < 0$, $\tau < 0$ | Complex, $\alpha < 0$ | Stable spiral |
| $\Delta > 0$, $\tau^2 - 4\Delta < 0$, $\tau > 0$ | Complex, $\alpha > 0$ | Unstable spiral |
| $\Delta > 0$, $\tau = 0$ | Pure imaginary | Center (stable) |

---

### 4.5 Nonhomogeneous Systems: Variation of Parameters (§4.4)

**System:** $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$

**Fundamental matrix:** $\Phi(t)$ — columns are linearly independent solutions of $\mathbf{x}' = A\mathbf{x}$.

**Particular solution:**

$$\mathbf{x}_p(t) = \Phi(t) \int \Phi(t)^{-1}\mathbf{f}(t)\, dt$$

**General solution:** $\mathbf{x}(t) = \Phi(t)\mathbf{c} + \mathbf{x}_p(t)$

---

## 5 Nonlinear Systems & Stability (Chapter 5)

### 5.1 Linearization at Equilibria (§5.1)

**System:** $\mathbf{x}' = \mathbf{F}(\mathbf{x})$, equilibrium at $\mathbf{x}^* = (x^*, y^*)$ where $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$.

**Jacobian matrix at $\mathbf{x}^*$:**

$$J = \begin{pmatrix} \partial F_1/\partial x & \partial F_1/\partial y \\ \partial F_2/\partial x & \partial F_2/\partial y \end{pmatrix}\Bigg|_{\mathbf{x}^*}$$

**Linearized system near $\mathbf{x}^*$:** $\mathbf{u}' = J\mathbf{u}$ where $\mathbf{u} = \mathbf{x} - \mathbf{x}^*$.

> **Note**
>
> The stability of $\mathbf{x}^*$ for the **nonlinear** system matches the Jacobian classification — **except** when the Jacobian gives a center ($\tau = 0$, $\Delta > 0$): the nonlinear system may be a center or a spiral. Further analysis (e.g., Lyapunov function) is needed.

**Steps:**

1. Find all equilibria: $F_1(x, y) = 0$ and $F_2(x, y) = 0$ simultaneously.
2. Compute $J$ and evaluate at each equilibrium.

3. Find eigenvalues of $J$; classify stability.
4. Sketch phase portrait near each equilibrium.

---

## 5.2 Stability Definitions (§5.1–§5.2)

| Type | Definition |
|---|---|
| **Stable** | Nearby trajectories stay nearby |
| **Asymptotically stable** | Nearby trajectories stay nearby **and** converge to $\mathbf{x}^*$ |
| **Unstable** | Trajectories move away from $\mathbf{x}^*$ |

---

## 5.3 Lyapunov Functions (§5.2)

**When to use:** Proving stability/instability without solving the system; especially useful when linearization is inconclusive (center case).

A function $V(\mathbf{x})$ is a **Lyapunov function** for $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ near $\mathbf{x}^*$ if:

- $V(\mathbf{x}^*) = 0$ and $V(\mathbf{x}) > 0$ for $\mathbf{x} \neq \mathbf{x}^*$ (positive definite).
- $\dot{V} = \nabla V \cdot \mathbf{F} \leq 0$ along trajectories.

| Condition on $\dot{V}$ | Conclusion |
|---|---|
| $\dot{V} \leq 0$ | $\mathbf{x}^*$ is **stable** |
| $\dot{V} < 0$ | $\mathbf{x}^*$ is **asymptotically stable** |
| $\dot{V} > 0$ (with $V(\mathbf{x}^*) = 0$) | $\mathbf{x}^*$ is **unstable** |

> **Tip**
>
> **Common Lyapunov candidates:** $V = x^2 + y^2$, $V = ax^2 + by^2$, energy functions $H(x, y)$.

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### 5.4 Limit Cycles & Periodic Orbits (§5.3)

**Limit cycle:** An isolated closed trajectory in the phase plane.

**Poincaré–Bendixson Theorem:** If a trajectory of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ in $\mathbb{R}^2$ remains in a bounded region containing no equilibrium, then it must approach a **closed orbit** (limit cycle).

**Bendixson's Criterion (no limit cycles):** If $\dfrac{\partial F_1}{\partial x} + \dfrac{\partial F_2}{\partial y}$ has constant sign in a simply connected region, then there are **no closed orbits** in that region.

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## 6 Overall Strategy Flowchart

```
Given an ODE/System:

  Single ODE, 1st order?
      Separable? → Separation of Variables
      Linear? → Integrating Factor
      Exact? → Exact Equations
      Bernoulli? → Bernoulli Substitution
      Homogeneous (y/x)? → Substitution v = y/x
      Autonomous? → Phase Line / Qualitative Analysis
      None of above → Numerical Methods (Euler / RK4)

  Single ODE, 2nd order, linear, const. coefficients?
      Homogeneous → Characteristic Equation
      Nonhomogeneous?
          g(t) = poly/exp/sin/cos → Undetermined Coefficients
          Other g(t) → Variation of Parameters

  Cauchy-Euler? → Substitution x = e^t

  Use Laplace Transform when:
      Initial conditions given
      Forcing is piecewise (step functions)
      Impulse forcing (Dirac delta)

  System of ODEs?
      Linear, const. coefficients → Eigenvalue Method
          Distinct real → Standard solution
          Complex → Euler's formula
          Repeated → Generalized eigenvectors
      Nonhomogeneous → Variation of Parameters
      Nonlinear → Linearize at equilibria (Jacobian)
          Stability unclear? → Lyapunov Functions
```

Quick Reminders

- Always verify your solution satisfies the original ODE.
- Apply initial/boundary conditions **after** finding the general solution.
- For systems, always check whether the phase portrait classification from the **trace-determinant plane** matches your computed eigenvalues.
- When using Laplace transforms, write the ODE in **standard form** before taking the transform.

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